

Executing and interpreting applications of PLS-SEM: Updates for family business researchers

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ABSTRACT

The use of partial least squares structural equation modeling (PLS-SEM) has been gaining momentum in family business research. Since the publication of a PLS-SEM guidelines article in the *Journal of Family Business Strategy*'s special issue on "Innovative and Established Research Methods in Family Business" in 2014, methodological research has developed new model evaluation methods and metrics and sharpened our understanding of the method's strengths and limitations. In light of these developments, we extend prior guidelines on PLS-SEM applications by discussing new model evaluation procedures (e.g., model selection) and metrics (e.g., PLS_{predict}). In addition, we highlight the usefulness of methodological extensions for discrete choice modeling and endogeneity assessment that considerably extend the scope of the PLS-SEM method, and emerging opportunities for the application of PLS-SEM with archival (secondary) data. PLS-SEM remains a valuable method in the context of family business research, especially when it comes to gaining a more sophisticated understanding of the drivers of family business behavior. Because of its properties, the PLS-SEM approach proves particularly valuable when the aim is to predict target variables (e.g., family firm performance) in the context of a causal model.

1. Introduction

Since its establishment in the early 1980s, the partial least squares (PLS) method has been a useful tool for researchers aspiring to analyze complex interrelationships between observed and latent variables (Wold, 1982). However, it was not until the *Journal of Family Business Strategy*'s dedicated 2014 special issue that the family business research community recognized the utility of the partial least squares structural equation modeling (PLS-SEM) in the family business context (Binz Astrachan, Patel, & Wanzenried, 2014; Sarstedt, Ringle, Smith, Reams, & Hair, 2014). In response to the special issue, family business researchers increasingly used PLS-SEM to explore phenomena pertaining

to both the business and the family domain. Pieper, Williams, Manley, and Matthews (2020), for example, find that among their sample of 105 US-based business families, those with more prevalent religious orientations favor long-term over short-term goals, and exhibit higher levels of family cohesion. Santiago, Pandey, and Manalac (2019), analyzing a sample of 252 non-professional investors in the Philippines, identify family presence as a driver of both family firm reputation and perceived financial performance. Using a sample of 125 German job applicants, Kahlert, Botero, and Prügl (2017) do not find a relationship between family ownership and organizational attractiveness. Instead, they confirm that organizational age is linked to applicants' compensation expectations. These three studies exemplify the applicability of PLS-SEM

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to assess the strength of complex relationships between constructs pertaining to the family and business domain, as well as the explanatory and predictive power of a model.

Methodological research on PLS-SEM has progressed considerably since the 2014 special issue. As a result, new model evaluation metrics have been developed that provide researchers with more options to more thoroughly analyze their models (e.g., Sarstedt et al., 2020). At the same time, conceptual articles discussing the properties of PLS-SEM have refined our understanding of the method's benefits and limitations (e.g., Rigdon, Sarstedt, & Becker, 2020). In light of these developments and family business researchers' ongoing interest in PLS-SEM, revisiting and extending prior guidelines seems timely and warranted.

In the 2014 lead article of the special issue – which, by mid-2020, has been cited more than 600 times in Google Scholar – Sarstedt, Ringle, Smith et al. (2014) introduce the PLS-SEM method to family business scholars, discuss its properties, and summarize the metrics for results evaluation. In this follow-up article, we summarize additional metrics to evaluate and report PLS-SEM results, with a particular focus on techniques and metrics recommended for family business researchers. As family business research is gaining traction in the general management literature, applying rigorous, cutting-edge methodological approaches becomes paramount. As an example, the increased interest in and understanding of the antecedents and effects of heterogeneity on both the family and business levels (Dibrell & Memili, 2019) call for methods that allow us to reliably generalize findings beyond the sample at hand (Nitzl & Chin, 2017) – that is, to both the family and non-family business populations. The combination of PLS-SEM's capability of delivering meaningful results, even on small samples, and the method's out-of-sample predictive power (Chin et al., 2020), makes the method particularly powerful and appropriate for the family business context.

This article is structured as follows. In the first section, we summarize our basic considerations in selecting PLS-SEM and the new metrics that have emerged. The second section describes advanced types of analyses that family business researchers should be aware of, so they can apply them as needed. We also summarize the advantages of PLS-SEM in the emerging era of research based on archival data. In closing, we envision methodological extensions required to further increase PLS-SEM's usefulness for family business researchers.

2. Basic PLS-SEM considerations and new metrics

In this section, we discuss the need to consider three important issues when applying PLS-SEM: (1) the *adequacy* of PLS-SEM, (2) assessment of *discriminant validity*, and (3) evaluating a model's *predictive power*. We strongly encourage family business scholars to consider the recent developments and apply these novel concepts and metrics when designing and reporting their research.

Considering the *first* issue, adequacy pertains to how and when to apply PLS-SEM, versus covariance-based SEM (CB-SEM). As decision criteria continue to evolve, we provide the reader with the most relevant considerations to guide their selection of the most appropriate SEM method. The *second* issue concerns the assessment of discriminant validity in PLS-SEM applications which, in the past, has largely relied on the Fornell and Larcker (1981) criterion. We highlight the advantages of the novel and more rigorous heterotrait-monotrait (HTMT) criterion (Henseler, Ringle, & Sarstedt, 2015) as a method to more accurately assess discriminant validity between constructs. The *third* issue introduces scholars to the latest methods for evaluating the predictive power of a model (Shmueli, Ray, Velasquez Estrada, & Chatla, 2016). While in the past reporting was often limited to metrics alluding to a model's explanatory power (i.e., in-sample predictive power), PLS_{predict} is a new tool that allows assessing a model's out-of-sample predictive power.

2.1. Adequacy: choosing between PLS-SEM and CB-SEM

The issue of whether to use PLS-SEM or CB-SEM has been widely recognized in recent years (Astrachan, Patel, & Wanzenried, 2014; Henseler et al., 2014; Sarstedt, Ringle, Smith et al., 2014). Several earlier studies compared the two methods without considering that they assume fundamentally different measurement philosophies (e.g., Goodhue, Lewis, & Thompson, 2012; Lu, Kwan, Thomas, & Cedzynski, 2011; Rönkkö & Evermann, 2013). Specifically, CB-SEM follows a common factor model, assuming that observed scores from the indicators are functions of the construct itself and of measurement error. To estimate the model parameters, such as indicator loadings and path coefficients, the method uses only common variance (i.e., variance shared by the indicators of each construct). By contrast, PLS-SEM follows a composite model logic, which uses total variance and represents constructs as linear combinations of its indicators (Sarstedt, Hair, Ringle, Thiele, & Gudergan, 2016).

Researchers have long recognized the differences between these paradigms, but only recently have discussions shifted towards understanding the implications of method choice (Hwang, Sarstedt, Cheah, & Ringle, 2020). Because of these conceptual differences, evaluating common factor models using composite-based PLS-SEM produces biased estimates. Similarly, estimating composite models with CB-SEM also produces biased estimates. In fact, Sarstedt et al. (2016) show that the biases produced by CB-SEM are far more severe than those of PLS-SEM, when applying the method to the wrong type of model (i.e., estimating composite models with CB-SEM vs. estimating common factor models with PLS-SEM).

In light of these conceptual differences, several researchers have called for the emancipation of composite-based methods from their covariance-based counterparts (e.g., Rigdon, 2012; Sarstedt, Ringle, Henseler, & Hair, 2014). Nonetheless, for many researchers SEM is equivalent to carrying out CB-SEM because of the common belief that concepts such as family influence (Astrachan, Klein, & Smyrnios, 2002) or family firm reputation (Binz, Hair, Pieper, & Baldauf, 2013) can only be measured using common factors, in which the indicator covariances define the nature of the data (Sarstedt et al., 2016). However, as Rigdon et al. (2017) describe, philosophy of science supports a broader perspective of measurement in that both approaches constitute approximations of the concept of interest. In fact, “common factor proxies cannot be assumed to carry greater significance than composite proxies in regard to the existence or nature of conceptual variables” (Rigdon, 2016, p. 602). Similarly, Rhemtulla, van Bork, and Borsboom (2020) observe that “there is a growing appreciation within some areas of psychology that the latent variable model may not be the right model to capture relations between many psychological constructs and their observed indicators.”

Furthermore, recent research has established the proxy nature of common factor models (Rigdon, Sarstedt, & Ringle, 2017; Sarstedt et al., 2016). Rigdon, Becker, and Sarstedt (2019) show that common factor models can be subject to considerable degrees of metrological uncertainty. Metrological uncertainty refers to the dispersion of the measurement values that can be attributed to the object/concept being measured (JCGM/WG1, 2008), which is well beyond the simple standard errors produced by CB-SEM analyses (Hair & Sarstedt, 2019). As such, uncertainty is a validity threat to measurement and has adverse consequences for the replicability of study findings (Rigdon et al., 2020). While uncertainty also applies to composite models, the way researchers treat models in CB-SEM analyses typically leads to a pronounced increase in uncertainty. More precisely, in an effort to improve model fit, researchers typically restrict the number of indicators per construct, which also increases uncertainty (Rigdon et al., 2019). These issues do not necessarily imply that composite models are superior, but they cast considerable doubt on the assumption of some researchers that CB-SEM constitutes the gold standard when measuring unobservable concepts. Instead, researchers should consider the entire toolbox of SEM methods,

including PLS-SEM and other composite-based methods, while acknowledging the proxy nature of their analyses.

2.2. Assessing discriminant validity using the HTMT statistic

A crucial aspect of measurement model evaluation involves the establishment of discriminant validity, which measures whether a construct is empirically different from other constructs in the same model (Farrell, 2010). The traditional and most widely applied metric in PLS-SEM analyses, until recently, was developed by Fornell and Larcker (1981). According to the Fornell-Larcker criterion, discriminant validity is established if, for each of two constructs, the squared multiple correlation between items and constructs (i.e., the average variance extracted, AVE) is greater than the shared variance between the constructs. On both empirical and conceptual grounds, however, recent research casts serious doubts upon the efficacy of this criterion (Franke & Sarstedt, 2019; Henseler et al., 2015).

As an alternative metric for discriminant validity assessment, Henseler et al. (2015) propose the heterotrait-monotrait (HTMT) ratio of correlations, the robustness of which has been confirmed in a series of follow-up studies (Franke & Sarstedt, 2019; Radomir & Moisescu, 2020; Voorhees, Brady, Calantone, & Ramirez, 2016). For any pair of latent variables, the HTMT is computed as the ratio between the mean value of the item correlations between constructs (i.e., the heterotrait correlations) and the (geometric) mean of the average correlations of the items measuring each construct (i.e., the two sets of monotrait correlations). Hence, a high HTMT value suggests discriminant validity problems between the two constructs. Henseler et al. (2015) suggest a cutoff value of 0.85 for two constructs to be considered as meeting the criteria for discriminant validity. However, if the constructs are conceptually similar, a higher, less conservative threshold value of 0.90 can be used. Examples of conceptually similar constructs are satisfaction and loyalty (e.g., Binz Astrachan, Prögl, Hair, & Babin, 2019), and trust and commitment (e.g., Smith, Hair, & Ferguson, 2014).

Franke and Sarstedt (2019) show that the application of cutoff values can conceal discriminant validity problems in the use of the HTMT statistic. Therefore, researchers should additionally use a statistical test to evaluate whether an HTMT value differs significantly from a selected threshold (e.g., 0.85). To test whether an HTMT value differs significantly from a threshold of say 0.85, researchers must construct the 95 % bootstrap-based confidence interval and examine whether the threshold falls within this interval. A bootstrap-based confidence interval that, in this example, includes 0.85 will indicate the HTMT does not differ significantly from this threshold value. Therefore, discriminant validity will not be established between the two corresponding constructs.

Recent research underlines that discriminant validity assessment requires particular attention when estimating higher-order constructs (Sarstedt, Hair, Cheah, Becker, & Ringle, 2019). A higher-order construct enables researchers to model a concept on a more abstract dimension (referred to as higher-order component) and its more concrete subdimensions (referred to as lower-order components). Each lower-order component is (typically) measured by multiple items and, hence, appears as a separate construct in a PLS path model. However, the reliability and validity assessment of a higher-order construct considers the lower-order components as indicators of the more abstract higher-order component. Thus, when assessing the discriminant validity of a reflectively measured higher-order construct, the heterotrait correlations correspond to the cross-loadings between the lower-order components of the higher-order construct and the indicators of all other reflective or single-item constructs in the model, while the monotrait correlations correspond to the correlations between the lower-order components of the higher-order construct.

For example, suppose that we want to examine the extent to which socioemotional wealth in family firms predicts risk propensity. In order to measure socioemotional wealth, we could employ a tri-dimensional measurement approach as suggested by Berrone, Cruz, and

Gomez-Mejia (2012), thereby assessing family-firm identification, emotional attachment to the firm, and transgenerational intent (i.e., desire to pass on to next generation) as reflective constructs, using a set of relevant items for each. Consequently, the three dimensions will be considered as the lower-order components (LOCs) of socioemotional wealth, which will be a reflectively measured higher-order construct (HOC). Furthermore, risk propensity, as an endogenous construct in the path model, can be measured reflectively using two indicators related to the firm's level of debt and acquisition propensity. Fig. 1 illustrates the computation of the HTMT value between the higher-order construct and the endogenous construct for this particular case. When assessing the discriminant validity of socioemotional wealth as an HOC, the heterotrait correlations will correspond to the cross-loadings of the risk propensity's indicators (i.e., level of debt and acquisition propensity) with the three LOCs of socioemotional wealth (i.e., family-firm identification, emotional attachment to the firm, and transgenerational intent). By contrast, monotrait correlations will correspond to the correlations among the three LOCs of socioemotional wealth and the two indicators of risk propensity.

As a cautionary note, before assessing an HOC's discriminant validity, researchers must also ensure that the LOCs meet the discriminant validity criteria, relative to one another as well as to the other reflectively measured, first-order constructs in the PLS path model. That is, in the example in Fig. 1, researchers also need to establish discriminant validity between risk propensity, family-firm identification, emotional attachment to the firm, and transgenerational intent.

2.3. Predictive power assessment using $PLS_{predict}$

Researchers typically seek to generalize their findings beyond the sample at hand (Nitzl & Chin, 2017). Producing generalizable findings requires assessing whether the results not only apply to the data that have been used in the model estimation process (in-sample), but also to other data sets (i.e., out-of-sample). Researchers using PLS-SEM (and regression-based methods in general) have typically neglected assessment of a model's out-of-sample predictive power. Instead, they primarily rely on the R^2 measure, which only indicates a model's explanatory power. However, the R^2 measure does not offer any evidence of the model's out-of-sample predictive power (Chin et al., 2020). This assessment requires the additional use of prediction-oriented evaluation criteria in PLS-SEM (Hair, 2020).

To assess a PLS path model's out-of-sample predictive power,

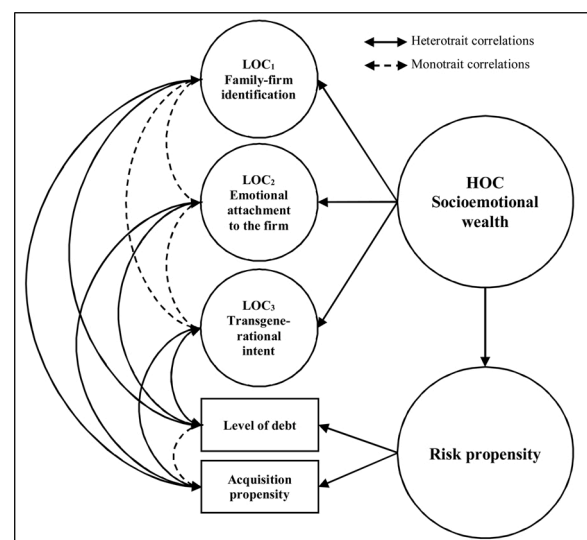


Fig. 1. HTMT conceptualization in the case of higher-order constructs (HOC = higher-order construct; LOC = lower-order construct).

researchers can apply PLS_{predict} (Shmueli et al., 2016), a holdout sample-based procedure that generates point predictions on the item as well as the construct level¹. PLS_{predict} divides the sample data into k subgroups (referred to as folds) of roughly the same size. The procedure then combines $k-1$ folds into a training sample that is used to estimate the model. The remaining fold serves as a holdout sample that is used to assess the model's predictive power. That is, the holdout sample includes those cases that will be predicted using the model parameters obtained from the estimation of the training sample. This process is repeated until each of the k folds has served as the holdout sample.

The default number of folds (k) is typically ten, but it can be adjusted upwards or downwards depending on the sample size. For example, if the total sample size is 200, using five folds would produce a sample size of 40 per fold. That is, the model would be estimated by removing one of the five folds from the sample observations ($200-40 = 160$), and the results would then be tested on the remaining 40 observations. Hence, if the model's minimum sample size requirement is higher than 160 (based on statistical power; Hair, Hult, Ringle, & Sarstedt, 2017), researchers need to choose a higher number of folds to obtain reliable parameter estimation.

The initial focus when interpreting PLS_{predict} results does not require an assessment of the prediction errors for all endogenous constructs. Rather, the primary concern is the prediction errors for the theoretical model's primary endogenous construct (assuming that the theoretical model includes more than a single endogenous construct). When the primary endogenous construct has been identified, the Q^2_{predict} statistic is examined first to determine whether the predictions are superior to the naïve benchmark criterion. This benchmark is calculated using the indicator means from the training sample and it is further used to predict the holdout sample (Shmueli et al., 2019). If the prediction results are better than the naïve benchmark criterion, as evidenced in Q^2_{predict} values above zero, then researchers need to examine further prediction statistics.

To further assess a model's predictive power, several prediction statistics are available that quantify the difference between the predicted and actual values (i.e., the prediction error). The most popular statistic is the root mean squared error (RMSE), which is defined as the square root of the average of the squared differences of the predictions and the actual values. Researchers typically compare the RMSE of the original model with a naïve benchmark criterion that assumes a simpler predictive mechanism. The recommended naïve criterion for these statistics uses a linear regression model (LM) to generate predictions for the manifest variables. For this purpose, a linear regression is run for each of the endogenous construct's indicators on the indicators of the exogenous latent variables in the PLS path model (Danks & Ray, 2018).

Shmueli et al. (2019) identify four possible outcomes when comparing the RMSE values from the PLS-SEM analysis with those produced by the naïve LM benchmark. Fig. 2 depicts these four outcomes.

In addition to PLS_{predict}, a more recent advancement of the prediction-oriented assessment of PLS-SEM results is the cross-validated predictive ability test (CVPAT; Liengard et al., 2020). PLS-SEM is the tool of choice when using SEM to build theories, if the statistical objective of the research is both explanation and prediction (Richter, Cepeda Carrión, Roldán, & Ringle, 2016; Shmueli, 2010). Until recently, a limitation of PLS-SEM was the lack of a statistical test to assess whether a proposed or alternative theoretical model achieves significantly better out-of-sample predictive power than any other established model. To overcome this limitation and to extend the causal-explanatory modeling nature of PLS-SEM and the prediction-oriented assessment of models, Liengard et al. (2020) developed the CVPAT. Different from

information-theoretic model selection criteria, the objective of which is to balance model fit and predictive power in the PLS-SEM context (Sharma, Sarstedt, Shmueli, Kim, & Thiele, 2019; Sharma, Shmueli, Sarstedt, Danks, & Ray, 2020), CVPAT focuses exclusively on a models' predictive power. Family business researchers can, accordingly, use the CVPAT to develop and validate theories that offer both explanation and prediction, and to select a model on which to base managerial and policy decisions. When applied, the test can strengthen the findings and conclusions of a PLS-SEM analysis, which often are of a prediction-oriented nature.

3. Advanced analysis: concepts and methods

A number of recent methodological developments have led to the addition of several advanced analysis methods that, although not yet mainstream, are likely to prove useful as specialized types of analyses when applied to future family business research. These developments are (1) the emergence of archival data, (2) the use of discrete choice modeling, and (3) the management of endogeneity issues.

3.1. Archival data and structural equation modeling

The data type used to examine relationships between variables and to predict outcomes, which is available for social sciences research, is increasingly shifting towards archival (secondary) data (Hair & Sarstedt, 2020). In comparison, most previous family business research relied on primary data from surveys. Survey-based research is typically measured reflectively, while archival data is often measured formatively since the attributes are measured in enterprise resource planning, customer relationship management, and similar data systems within a business organization. For example, researchers may be working with social media and website interaction data, measuring different types of customer interactions, inventory management, production capacity, accounting and transactions, purchase orders, compensation, and so forth, all of which can be considered formative. Whereas PLS-SEM readily accommodates formatively specified constructs, CB-SEM does not. This makes PLS-SEM a more suitable analysis method to estimate this type of data.

When the aim is to confirm well-developed theoretical models, researchers typically rely on survey data. By contrast, archival data applications are often used in research that is more exploratory in nature, or when the aim is to predict relevant outcome variables such as business performance (Hair, Sarstedt, & Ringle, 2019). If the focus is on prediction and the ability to assess a model's predictive power, PLS-SEM perfectly meets these requirements (Jöreskog & Wold, 1982; Shmueli et al., 2019). We expect the applications of PLS-SEM to further accelerate, as the use of archival data becomes more widespread.

3.2. PLS-SEM using discrete choice experiment data

In family business research, discrete choice modeling and related techniques (e.g., choice-based conjoint) are useful in several decision-making contexts, enabling family business researchers to explain and predict choices between two alternatives. Discrete choice modeling is often implemented as part of experiments that systematically manipulate conditions to assess causality (Lude & Prügl, 2020). With the growing popularity of experiments in family business research, discrete choice modeling occupies an increasingly important role in the field.

In applying discrete choice modeling, researchers attempt to determine the benefits (value or utility) an individual has in choosing a particular alternative. In the family business context, previous applications include family business succession (Mathews & Blumentritt, 2015), private equity investment decisions (Dawson, 2011), and attracting new talent to family firms (Hauswald, Hack, Kellermanns, & Patzelt, 2016). These research approaches use binary or dummy-coded (i.e., Boolean) variables to capture respondents' choice behavior.

¹ Shmueli et al. (2019) offer detailed guidelines for running the PLS_{predict} procedure and interpreting its results.

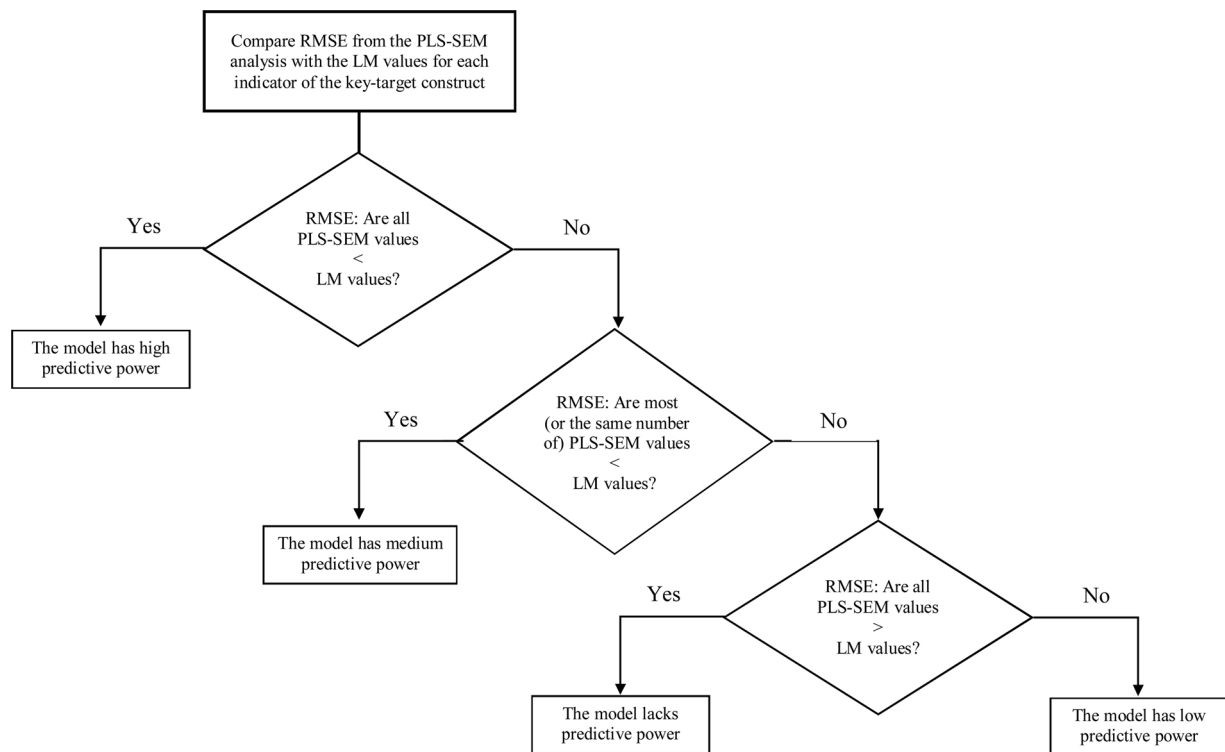


Fig. 2. Interpretation of $PLS_{predict}$ results.

Researchers can include these choice variables in PLS path models although PLS-SEM generally requires metric or quasi-metric indicator data (Hair et al., 2017). A recent publication by Hair, Ringle et al. (2019) shows how to run PLS-SEM with data from a discrete choice experiment (Louviere, Hensher, & Swait, 2000). The data stem from experimental designs and have the advantage that different levels of a particular attribute appear equally frequently in the data set and are orthogonal. Discrete choice modeling typically relies on multinomial logit or linear probability models for model estimation. Estimating a PLS path model by using discrete choice experiment data follows a four-step procedure, which produces similar results like those in conditional logit model estimation. Also, the structural model estimates produced by PLS-SEM match the importance weights indicating the relevance of attributes as a whole – as typically determined in discrete choice modeling through subsequent calculations. In light of the flexibility of PLS-SEM with discrete choice experiment data, we anticipate that family business researchers will be likely to explore the potential of combining experimental research designs with PLS-SEM. Indeed, this approach is suitable for examining decision making in the whole spectrum of family business research.

3.3. Endogeneity

Review studies on the use of PLS-SEM show that researchers seldom address endogeneity in their PLS-SEM applications (e.g., Nitzl & Chin, 2017) – and family business research is no exception (Sarstedt, Ringle, Smith et al., 2014). This could be the result of the prediction-oriented character of PLS-SEM (Cepeda Carrión, Henseler, Ringle, & Roldán, 2016; Evermann & Tate, 2016). Endogeneity is a situation in which an independent variable in a statistical model, such as a PLS path model, is correlated with the dependent variable's error term. If the objective of model estimation is fully predictive, endogeneity is not an issue (Ebbes, Papies, & van Heerde, 2016). However, in explanatory models with causal relationships the exact estimation of the coefficients is desirable, and researchers should ensure that endogeneity does not substantially impact their results (Park & Gupta, 2012). When researchers apply

PLS-SEM as a causal-predictive approach (Sarstedt, Ringle, & Hair, 2017), endogeneity is a relevant issue (Hult et al., 2018), especially when the research entails both explanatory and predictive goals (Richter et al., 2016; Shmueli, 2010). This also applies to research on family businesses (Sarstedt, Ringle, Smith et al., 2014).

Endogeneity can have many sources, such as measurement error or simultaneity (Ebbes et al., 2016; Hult et al., 2018). But endogeneity often results from omitted variables. This means researchers fail to identify all the relevant independent variables to explain the dependent variable. For example, in their study, Basco, Calabrò, and Campopiano (2019) examine the impact of several dimensions of corporate entrepreneurship and “familiness” on transgenerational entrepreneurship. If the authors only consider corporate entrepreneurship as an antecedent to transgenerational entrepreneurship, their model will likely suffer from pronounced endogeneity due to the omission of “familiness.”

To address endogeneity problems that result from omitted variables, researchers can choose from a variety of approaches, most of which are implemented post hoc. The most prominent approaches are the use of control variables (e.g., Ebbes et al., 2016) and instrumental variables (IVs; e.g., Bascole, 2008). A major problem of these techniques, however, is that theoretically plausible control variables and IVs are difficult to identify or are simply not available. Using inappropriate IVs can even increase the bias (i.e., the cure can be more dangerous than the disease; Ebbes et al., 2016). When using IVs in applications, researchers cannot verify if the results improved or worsened.

Gaussian copulas represent a more recent approach to address endogeneity issues (Park & Gupta, 2012). The main advantage of this method is that it does not require researchers to identify suitable IV or control variables. In addition, the Gaussian copula approach not only allows the identification of endogeneity issues, but also their handling. Hult et al. (2018) introduce this approach to PLS-SEM and offer guidelines to identify and treat endogeneity in PLS path models.

4. Discussion and conclusions

PLS-SEM was specifically designed to explore the relationships

among abstract concepts, which makes it an attractive choice of method across disciplines – the domain of family business being no exception. For instance, in the family business context, complex mechanisms such as family cohesion (Pieper, 2010) have been shown to be important drivers of family business behavior. However, their conceptualization, operationalization, and measurement has proven to be difficult due to the limitations of more traditional methodological approaches.

The PLS-SEM method, particularly in light of some of the new methodological extensions described in this article (Hwang et al., 2020; Khan et al., 2019), is a “good fit” for family business scholars seeking to investigate relationships between abstract concepts, especially when exploring success factors and the sources of competitive advantage (Hair, Sarstedt, Pieper, & Ringle, 2012). Furthermore, PLS-SEM is particularly useful when it comes to including mediators in the structural model. For example, Sarstedt, Hair, Nitzl, Ringle, and Howard (2020) emphasize the superiority of PLS-SEM for mediation and conditional process analyses (i.e., analyses that combine mediation and moderation) over regression analyses using Hayes’s (2018) PROCESS macro, which also features prominently in family business research (e.g., van Gils, Huybrechts, Minola, & Cassia, 2019). Contrary to PROCESS-based analyses, PLS-SEM treats the effect chain, as defined by the structural model relationships, as a holistic process and accounts for the measurement error inherent in measures of abstract concepts. In another extension, Richter, Schubring, Hauff, Ringle, and Sarstedt (2020) subject the composite scores produced by PLS-SEM to a necessary condition analysis (NCA). NCA differs from PLS-SEM, which follows a sufficiency logic (“X increases Y”). By following a necessity logic, NCA implies that an outcome – or a certain level of an outcome – can only be achieved if the necessary cause is in place, or is at a certain level (“X is needed for Y”). As an example of a necessity logic analysis, González-Cruz and Cruz-Ros (2016) apply a method similar to NCA. In their research they investigate the required conditions and levels of family involvement in the family business context – a key expression of family business heterogeneity (e.g., Chua, Chrisman, Steier, & Rau, 2012) – to produce superior performance in small and medium-sized enterprises, and family businesses.

If the goal is indeed to deepen our understanding of how and why family firms differ from one another, as well as from firms lacking family influence, we need to acknowledge and investigate the numerous sources of heterogeneity in both the family (e.g., levels of family involvement) and the business (e.g., debt ratio) domains, as well as their effect on family (e.g., family cohesion) and business (e.g., performance) outcomes (Dibrell & Memili, 2019). Interestingly, prior research almost exclusively focused on expressions of heterogeneity in the business domain (e.g., number of family members on the board), and the way in which they affect business outcomes, such as performance (e.g., Basco, Campopiano, Calabrò, & Kraus, 2019; García-Ramos & García-Olalla, 2011) or innovation expenditures (Arzubiaga, Kotlar, De Massis, Maseda, & Iturralde, 2018; Matzler, Veider, Hautz, & Stadler, 2015). While these observations are important to further our understanding of family business behavior, they omit the fact that observable behaviors are likely an expression of largely unexplored family processes. Hence, not only in this research domain but throughout the management sciences, replication studies and analyses of changes across time increase in relevance.

As an example, let us assume our rigorous research clearly shows that larger shareholder groups run more profitable companies, leading to a causal relationship between shareholder group size and performance. This causal relationship is most likely incomplete, however, since it omits any prior family processes that led to a larger shareholder base in the first place. Highly cohesive, multigenerational families are far less likely to “prune the tree,” given their ability to absorb and manage conflict, their commitment to one another and to the business, and their shared goals that go beyond being together in business. Hence, the observable performance effects are only indirectly rooted in the larger shareholder base, whose commitment and talents can be leveraged for

the sake of the business. Family firms with a larger shareholder base outperform others, therefore, because of the underlying family dynamics that allow such observable behaviors. Hence, if we report the findings of our incomplete causal chains, we may be doing the family business practitioner community a disservice, as expanding the shareholder base will likely exacerbate the conflict for a family with low cohesion.

PLS-SEM and its novel methodological extensions represent a valuable instrument to investigate complex path relationships to disclose the causal chains between constructs of interest (Ringle, Sarstedt, Mitchell, & Gudergan, 2020). For example, PLS-SEM-based endogeneity assessment allows family business researchers to correct biases in the model estimates that result from omitted variables (Hult et al., 2018). Guidelines for running mediation analyses offer family business researchers concrete guidance on how to analyze indirect effects among multiple constructs in highly complex model set-ups (Nitzl, Roldán, & Cepeda Carrión, 2016; Sarstedt, Hair et al., 2020). Updated guidelines for PLS-SEM-based model assessment (e.g., Hair, Risher, Sarstedt, & Ringle, 2019; Hair, Howard, & Nitzl, 2020; Sarstedt et al., 2020) consider and include recent developments in validity assessment, including approaches dealing with (unobserved) heterogeneity (e.g., Hair, Sarstedt, Matthews, & Ringle, 2016; Matthews, Sarstedt, Hair, & Ringle, 2016; Sarstedt, Ringle, & Hair, 2017).

These extensions not only facilitate a more holistic assessment of research results (e.g., regarding the model’s predictive power), but also enable researchers to consider new research contexts in their PLS-SEM analyses (e.g., by using data from discrete choice experiments). Researchers in the area of family business, especially those endeavoring to advance their field by applying the methods that best fit their research context and objectives, should consider these extensions – also considering that reviewers and editors increasingly require authors to meet the highest methodological standards. In fact, methodological rigor has become a sine qua non in top-tier family business research journals (Wilson et al., 2014) and we expect this trend to continue.

Recent methodological research has carved out various situations that favor the use of PLS-SEM, and future methodological research on PLS-SEM will increase the method’s capabilities. For example, extensions of the modeling options (e.g., simultaneous, and constrained relationships) can increase the method’s flexibility, and thereby also its usefulness for researchers to create and estimate their theoretical models. Also, future advances of the PLS-SEM method can accommodate multi-level and longitudinal data, as well as Bayesian approaches. Finally, we expect additional improvements of methods and criteria that further support the explanatory and predictive assessment of PLS-SEM results.

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